Yashasvi Nagar

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PROFESSIONAL SUMMARY

Graduate student in Information Systems at Northeastern University with hands-on experience building LLM-powered agentic AI systems, data pipelines, and knowledge graphs. Skilled in Python, SQL, and cloud platforms (Azure, Databricks, Snowflake), with end-to-end experience from raw data ingestion to production ML models. Passionate about applying LLMs and RAG to extract knowledge from complex engineering data and accelerate mission-critical work.

EDUCATION

Northeastern University | *Master of Science, Information Systems* **Dec 2026**

Acropolis Institute of Technology and Research | *B. Tech, Computer Science and Engineering* **Jun 2024**

SKILLS

Programming Languages: Python, SQL, R
Frameworks & Libraries: BERT, DialogueRNN, PySpark, Streamlit, dbt, NetworkX, OpenCV, GRU
Databases: Neo4j, SQL Server, Snowflake
Tools & Software: Databricks, Azure Data Factory, Tableau, Power BI, Alteryx Designer, ER/Studio, EDA, Agile/Scrum, Data Governance
Cloud Platforms: Microsoft Azure, GCP

WORK EXPERIENCE

Cernuvo | *AI Full-Stack Engineer Co-op* **Jan 2026 - Jun 2026**

- Developed GraphRAG-based features for an AI-powered web application to enable context-aware knowledge retrieval.
- Designed and optimized graph data models to support scalable querying and efficient information linking.
- Collaborated with the Technical Co-Founder on feature design, user experience flows, and system improvements.
- Built, tested, and deployed full-stack AI features while writing clean, maintainable, production-ready code.

Cummins Turbo Technologies | *Intern* **Oct 2023 - Feb 2024**

- Worked on a live project focused on image processing and computer vision using Python and OpenCV.
- Collaborated with cross-functional teams to enhance real-time processing performance and optimize system efficiency.
- Implemented object detection and image analysis features using OpenCV libraries in Python.
- Followed Agile methodology for iterative development, sprint planning, and regular project updates.

PROJECTS

PriorAuthPilot **Sep 2026 - Present**

- Building a RAG pipeline over NCD/LCD policy documents from the Medicare Coverage Database with metadata-filtered retrieval by payer, procedure, and jurisdiction, eliminating cross-policy mismatches before semantic search runs
- Designing a LangGraph-based agentic workflow with conditional branching that routes prior authorization cases through retrieval, criteria evaluation, and submission based on documentation completeness and denial risk
- Developing deterministic rule-evaluation logic to check patient records against structured policy criteria, replacing opaque LLM judgment with auditable, testable coverage checks
- Implementing three MCP servers wrapping distinct healthcare systems - Claims & Patient Data, Policy Knowledge (RAG), and Submission & Tracking - with an appeal-drafting module that parses CARC/RARC denial codes to generate evidence-grounded responses

MediQuery **Jul 2026 - Sep 2026**

- Clinical Analytics Platform with Safety-First AI
- Building an end-to-end healthcare data pipeline processing synthetic FHIR patient records through Medallion architecture with dbt transformations across bronze, silver, and gold layers.
- Designing a safety-first GraphRAG agent on Neo4j with mandatory citation guards and confidence scoring, rejecting any AI response not grounded in source FHIR resources.
- Implementing multi-tier role-based access control (Doctor, Researcher, Admin, Patient) and a natural-language cohort builder for clinical queries.

CrimeGraphRAG **Sep 2025 - Dec 2025**

- AI-Powered Crime Investigation System
- Developed an LLM-powered investigation agent using Neo4j and GraphRAG with reduced hallucination.
- Built complex knowledge graph modeling crimes, suspects, locations, organizations, and evidence.
- Implemented NetworkX visualizations in Streamlit for exploring criminal connections and patterns.
- Designed a conversational AI dashboard with real-time stats and role-based access control for multiple user roles.

Emotion Recognition in Multi-Turn Conversations **Jul 2025 - Aug 2026**

- Developed context-aware emotion detection on the MELD dataset (13,708 utterances, 1,433 conversations) for 7-class classification across multi-speaker dialogues.
- Compared BERT baseline, BERT + self-attention, and DialogueRNN architectures - DialogueRNN achieved 70.4% accuracy and 71.2% weighted F1, an 11.3% accuracy gain over baseline by explicitly modeling speaker states and conversation flow with GRU networks.
- Addressed severe class imbalance (Neutral 38% vs Fear 2.4%) with inverse-frequency weighted cross-entropy loss, stratified sampling, and gradient accumulation to simulate larger batch sizes under memory constraints.
- Built a Streamlit web app supporting single-utterance and multi-turn conversation analysis with real-time predictions, speaker-wise emotion distributions, and confidence score visualizations.